INCH-POUND

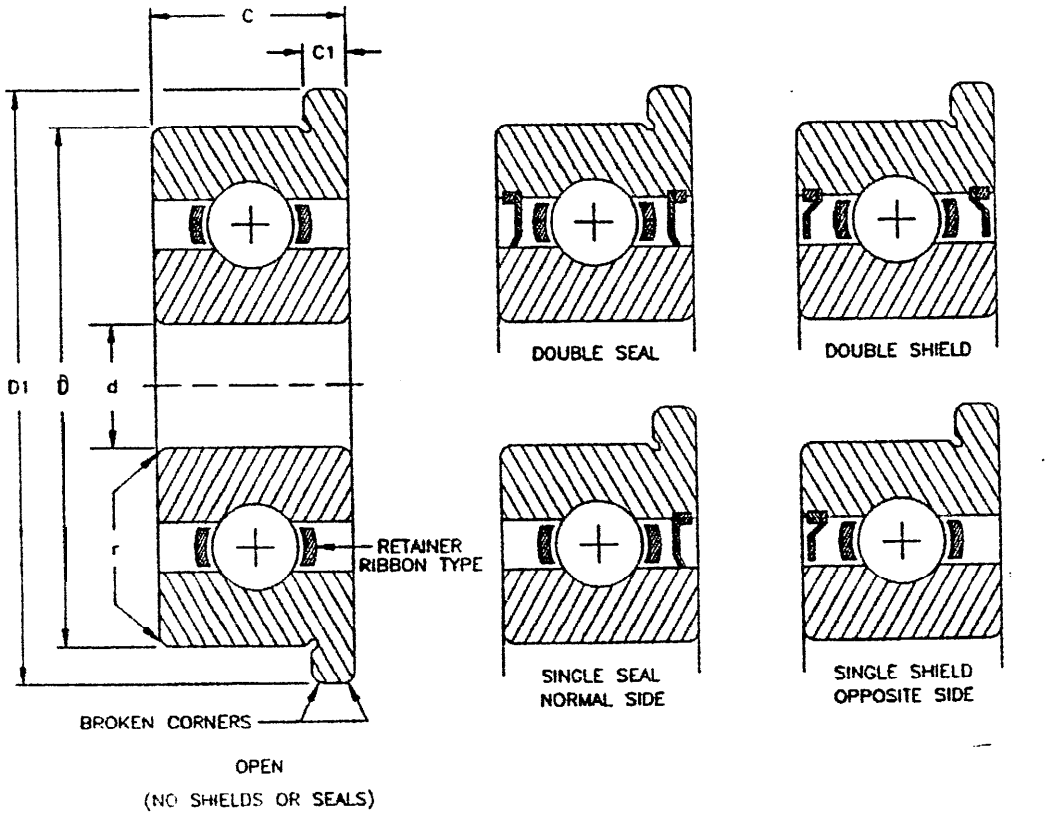
MIL-B-81793/8
15 MARCH 1993

MILITARY SPECIFICATION SHEET

# BEARING, BALL, ANNULAR, FOR INSTRUMENTS AND PRECISION ROTATING COMPONENTS, DEEP GROOVE, FLANGED, CORROSION RESISTANT STEEL, ABEC 7P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-81793




BEARING CONFIGURATIONS

AMSC N/A

FSC 3110

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REQUIREMENTS

1. Design. All bearings described by this specification sheet shall be deep groove instrument bearings, flanged.

2. Material. The ball and ring material for these bearings shall be corrosion resistant steel 440C (UNS S44004) conforming to QQ-S-763.

3. Tolerance class. The tolerance class shall be in accordance with the ABEC 7P tolerance tables of ANSI/AFBMA Standard 12.2. This tolerance class shall apply to all bearing sizes listed in Table I of this specification sheet.

4. Boundary dimensions. The boundary dimensions shall be as specified by the dash number (DASH NO) in Table I.

5. Retainer. The retainer shall be as specified by the part number designator (PN DES) in Table II.

6. Closures. Closures shall be as specified by the part number designator in Table III.

7. Radial internal clearance. The radial internal clearance shall be as specified by the part number designator in Table IV.

8. Calibration. Calibration shall be as specified by the part number designator in Table V.

9. Performance Test. The performance test shall be as specified by the part number designator in Table VI.

10. Lubrication.

10.1. Lubricant. The lubricant shall be in accordance with the specification specified by the part number designator in Table VII.

10.2 Lubricant amount. The amount of lubricant shall be as specified by the part number designator in Table VIII.

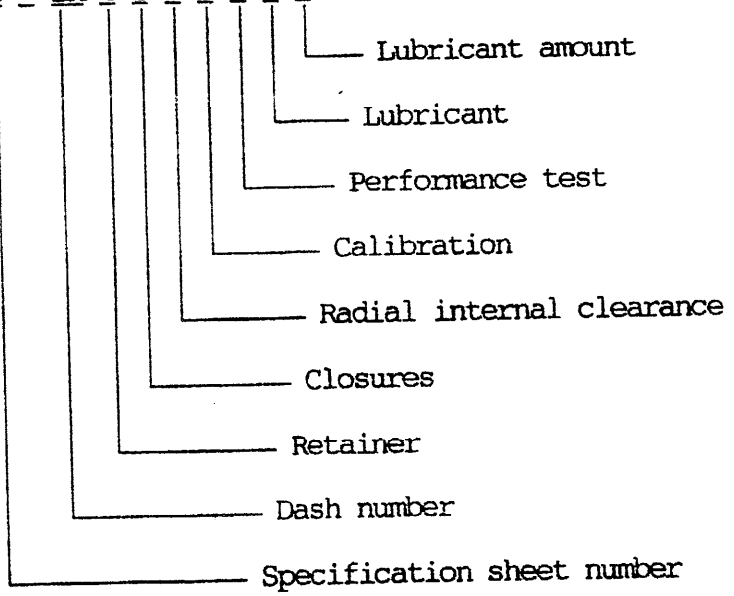
10.3. Barrier coating. Barrier coating shall be applied to bearings as specified by the part number designator in Table WI.



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11. Part number. The part number consists of the following: prefix letter M; general specification number; number of this specification sheet; dash number; and characters for retainer, closures, radial internal clearance, calibration, performance test, lubricant, and lubricant amount.

EXAMPLE: M 81793/ 08 - BA 4 C 2 N 2 A 1




M81793/08-BA4C2N2A1 indicates - Bore 0.0469"; outside diameter 0.1562"; width 0.0625"; flange OD 0.203"; flange width 0.013"; radius 0.003; retainer, ribbon, loose clinched; closures, 2 shields; radial internal clearance, 0.0002" to 0.0005"; no calibration; performance test, starting torque - Grade A; lubricant, MIL-L-6085; lubricant amount, manufacturer's standard.



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TABLE I. Boundary dimensions (inches).

| DASH NO | BORE | OD | WIDTH | FLANGE OD | FLANGE WIDTH | RADIUS |
|---|---|---|---|---|---|---|
| | d | D | C | $D_1$ | $C_1$ | r1/ |
| -AA | 0.0400 | 0.1250 | 0.0469 | 0.171 | 0.013 | 0.003 |
| -BA | 0.0469 | 0.1562 | 0.0625 | 0.203 | 0.013 | 0.003 |
| -BB | 0.0469 | 0.1562 | 0.0937 | 0.203 | 0.031 | 0.003 |
| -CA | 0.0550 | 0.1875 | 0.0781 | 0.234 | 0.023 | 0.003 |
| -CB | 0.0550 | 0.1875 | 0.1094 | 0.234 | 0.031 | 0.003 |
| -DA | 0.0781 | 0.2500 | 0.0937 | 0.296 | 0.023 | 0.003 |
| -DB | 0.0781 | 0.2500 | 0.1406 | 0.296 | 0.031 | 0.003 |
| -EA | 0.0937 | 0.1875 | 0.0625 | 0.234 | 0.018 | 0.003 |
| -EB | 0.0937 | 0.1875 | 0.0937 | 0.234 | 0.031 | 0.003 |
| -EC | 0.0937 | 0.3125 | 0.1094 | 0.359 | 0.023 | 0.003 |
| -ED | 0.0937 | 0.3125 | 0.1406 | 0.359 | 0.031 | 0.003 |
| -FA | 0.1250 | 0.2500 | 0.0937 | 0.296 | 0.023 | 0.003 |
| -FB | 0.1250 | 0.2500 | 0.1094 | 0.296 | 0.031 | 0.003 |
| -FC | 0.1250 | 0.3125 | 0.1094 | 0.359 | 0.023 | 0.003 |
| -FD | 0.1250 | 0.3125 | 0.1406 | 0.359 | 0.031 | 0.003 |
| -FE | 0.1250 | 0.3750 | 0.1094 | 0.422 | 0.023 | 0.005 |
| -FF | 0.1250 | 0.3750 | 0.1406 | 0.422 | 0.031 | 0.005 |
| -FG | 0.1250 | 0.3750 | 0.1562 | 0.440 | 0.030 | 0.012 |
| -GA | 0.1562 | 0.3125 | 0.1094 | 0.359 | 0.023 | 0.003 |
| -GB | 0.1562 | 0.3125 | 0.1250 | 0.359 | 0.036 | 0.003 |
| -GC | 0.1562 | 0.3125 | 0.1094 | 0.359 | 0.023 | 0.003 |
| -HA | 0.1875 | 0.3750 | 0.1250 | 0.422 | 0.023 | 0.003 |
| -HB | 0.1875 | 0.3750 | 0.1250 | 0.422 | 0.031 | 0.003 |
| -HC | 0.1875 | 0.5000 | 0.1562 | 0.565 | 0.042 | 0.012 |
| -HD | 0.1875 | 0.5000 | 0.1960 | 0.565 | 0.042 | 0.012 |
| -JA | 0.2500 | 0.3750 | 0.1250 | 0.422 | 0.023 | 0.003 |
| -JB | 0.2500 | 0.3750 | 0.1250 | 0.422 | 0.036 | 0.003 |
| -JC | 0.2500 | 0.5000 | 0.1250 | 0.547 | 0.023 | 0.005 |
| -JD | 0.2500 | 0.5000 | 0.1875 | 0.547 | 0.045 | 0.005 |
| -JE | 0.2500 | 0.6250 | 0.1960 | 0.690 | 0.042 | 0.012 |
| -KA | 0.3125 | 0.5000 | 0.1562 | 0.547 | 0.031 | 0.005 |
| -LA | 0.3750 | 0.8750 | 0.2188 | 0.969 | 0.062 | 0.016 |
| -LB | 0.3750 | 0.8750 | 0.2812 | 0.969 | 0.062 | 0.016 |
| -MA | 0.5000 | 1.1250 | 0.2500 | 1.225 | 0.062 | 0.016 |
| -MB | 0.5000 | 1.1250 | 0.3125 | 1.225 | 0.062 | 0.016 |

1/ Maximum shaft or housing fillet radius that bearing corners will clear.



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TABLE II. Retainer.

| PN DES | TYPE |
|---|---|
| 1 | MANUFACTURER'S STANDARD 1/ |
| 2 | CROWN 2/ |
| 3 | RIBBON, TIGHT CLINCHED 3/ |
| 4 | RIBBON, LOOSE CLINCHED 3/ |
| 5 | PTFE TUBE SEPARATOR 4/ |
| 6 | NOT USED |
| 7 | PHENOLIC LAMINATE 5/, 7/ |
| 8 | VACUUM IMPREGNATED PHENOLIC 6/, 7/ |
| 9 | NON-POROUS, NON-METALLIC CROWN 7/ |

1/ One piece pressed corrosion resistant steel crown, or two piece pressed corrosion resistant steel ribbon.
2/ One piece pressed corrosion resistant steel.
3/ Two piece pressed corrosion resistant steel.
4/ PTFE (Polytetrafluoroethylene).
5/ Phenolic or other porous non-metallic material.
6/ Phenolic or other porous non-metallic material saturated with lubricant. Selection of this choice dictates choosing manufacturer's standard amount of lubricant.
7/ Used for high speed applications.

TABLE III. Closures.

| PN DES | NUMBER | TYPE | LOCATIONS |
|---|---|---|---|
| N | NONE | NONE | |
| A | ONE | SHIELD 1/ | NORMAL 4/ |
| B | ONE | SHIELD 1/ | OPPOSITE 5/ |
| C | TWO | SHIELD 1/ | |
| D | ONE | SEAL 2/ | NORMAL 4/ |
| E | ONE | SEAL 3/ | NORMAL 4/ |
| F | ONE | SEAL 2/ | OPPOSITE 5/ |
| G | ONE | SEAL 3/ | OPPOSITE 5/ |
| H | TWO | SEAL 2/ | |
| J | TWO | SEAL 3/ | |

1/ Corrosion resistant steel conforming to ASTM-A-580, Condition A and ASTM-A-313, Condition C or QQ-S-766.
2/ Synthetic rubber.
3/ Glass fiber (Polytetrafluoroethylene) or other inert fiber.
4/ On flange side.
5/ On side opposite flange.



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TABLE IV. Radial Internal Clearance.

| PN DES | RANGES (inches) |
|---|---|
| 1 | 0.0001 to 0.0003 |
| 2 | 0.0002 to 0.0005 |
| 5 | 0.0005 to 0.0008 |
| 8 | 0.0008 to 0.0011 |

TABLE V. Calibration of bore and outside diameter.

| PN DES | BORE INCREMENTS | OD INCREMENTS |
|---|---|---|
| N | NO CALIBRATION | NO CALIBRATION |
| A | NO CALIBRATION | 0.00010 |
| B | NO CALIBRATION | 0.00005 |
| C | 0.00010 | NO CALIBRATION |
| D | 0.00010 | 0.00010 |
| E | 0.00010 | 0.00005 |
| F | 0.00005 | NO CALIBRATION |
| G | 0.00005 | 0.00010 |
| H | 0.00005 | 0.00005 |

TABLE VI. Performance test.

| PN DES | TYPE |
|---|---|
| 1 | MANUFACTURER'S STANDARD |
| 2 | STARTING TORQUE 1/ |

1/ Starting torque limits from table III of MIL-B-81793.



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TABLE VII. Lubricant.

| PN DES | SPECIFICATION |
|---|---|
| P | PRESERVATIVE 1/ |
| A | MIL-L-6085 |
| B | MIL-L-6085 2/ |
| C | MIL-L-81846 |
| D | MIL-L-81846 2/ |
| E | MIL-G-23827 |
| F | MIL-G-81322 |
| G | MIL-G-81937 |
| H | MIL-G-83261 |
| J | MIL-S-81087 |
| K | MIL-S-81087 2/ |

1/ PN DES "P" shall be used only with PN DES "P" in TABLE VIII.
2/ With barrier coat.

TABLE VIII. Lubricant amount.

| PN DES | QUANTITY |
|---|---|
| P | PRESERVATIVE AMOUNT AS REQUIRED BY MIL-P-197 |
| 1 | MANUFACTURER'S STANDARD 1/ |
| 2 | OIL PER TABLE II of MIL-B-81793 |
| 3 | 15% GREASE PER TABLE II of MIL-B-81793 |
| 4 | 25% GREASE PER TABLE II of MIL-B-81793 |
| 5 | 35% GREASE PER TABLE II of MIL-B-81793 |
| 6 | 45% GREASE PER TABLE II of MIL-B-81793 |

1/ Grease, fill to minimum 25%, maximum 40% of bearing void. Oil, immerse and then allow excess to drip off. The standard quantity of oil varies with each bearing size.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
Navy - SH
DLA - IS

User activities:
Navy - MC
Air Force - 84

Preparing activity:
Navy - AS

Agent:
DLS - IS

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